Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist *July 2022 - Present*
Nevada National Security Site

- Used Pytorch and Scikit-image to create a model of an X-ray machine's source, which was deployed to the engineering team to deconvolve and deblur X-ray images, that aided in machine configuration
- Trained a Keras model consisting of LSTM and dense layers to forecast future performance of an X-ray machine with an accuracy of 90%, which improved maintenance planning and machine repair time
- Developed interactive dashboards using Streamlit and Panel to visually present complex data and provide actionable insights to internal engineers and stakeholders
- Lead investigator for a data engineering project involving the organization and feature extraction of a dataset consisting of 12 years of linear accelerator data from the Los Alamos National Lab
- Maintainer of a company-wide seismic data-processing Python library hosted on our internal GitLab
- Mentored a student summer-intern and conducted weekly code review meetings to ensure best practices in programming
- Built and managed a MySQL database of seismic sensor metadata

Graduate Teaching Assistant *January 2016 - June 2022*
University of Nevada, Las Vegas and Auburn University

- Instructed sections in Mathematical Statistics, Real Analysis, Calculus I, II and III and Pre-Calculus
- Managed Auburn University's math tutoring center of 14 tutors

Technical Skills

Programming Languages: Python, SQL, Cypher, Bash, R, Lua, LaTeX

Libraries: Pytorch, Scikit-Learn, Scipy, Pandas, Sqlalchemy, Plotly, Panel

Developer Tools: Git, AWS, Unix/Linux, Vim, Tmux, Neo4j

Education

Auburn University - Ph.D. *June 2022*
Research Area: Probability (Stochastic Processes and Partial Differential Equations)
Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S. *December 2015 / January 2018*
Major: Applied Mathematics

Publications

1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). https://doi.org/10.1007/s40072-022-00258-6

2: Eisenberg, N., Chen, L. Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ with no drift term. Accepted November 2023 to Journal of Theoretical Probability.
Preprint available at https://arxiv.org/pdf/2209.04771v1.pdf